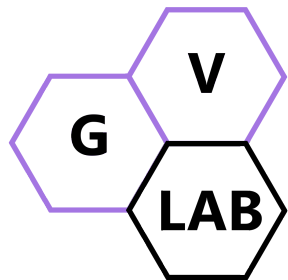


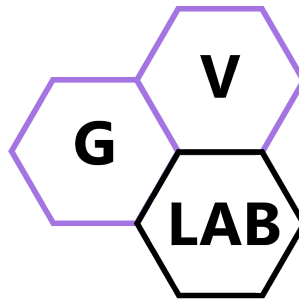
Convolution and Cross-correlation

Dr. Thanh-Sach LE
LTSACH@hcmut.edu.vn



GVLab:
Graphics and Vision Laboratory

**Faculty of Computer Science and Engineering,
HCMUT**

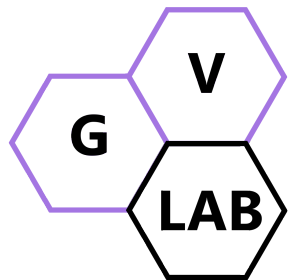


- ❖ What & Why?
- ❖ Mathematical Definition
- ❖ Computation of convolution

Convolution and Cross-correlation

What & Why?

Dr. Thanh-Sach LE
LTSACH@hcmut.edu.vn



GVLab:
Graphics and Vision Laboratory

Faculty of Computer Science and Engineering,
HCMUT

- ❖ Convolution and cross-correlation
 - ❖ important operations in signal processing
 - ❖ used to transform input signal (e.g. image) to output feature map and from input feature map to output feature map
 - ❖ hence, in deep neural network they are used to learn (to extract) features from input signals
- ❖ Convolution in Deep Learning
 - ❖ Convolution is widely used to design deep neural networks (see following slides)

discrete distribution for 1000 classes

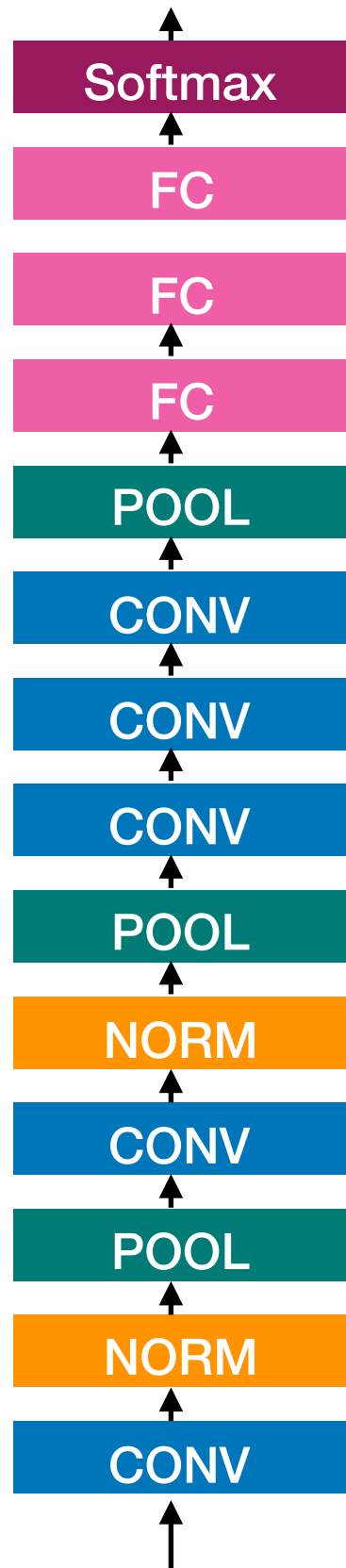
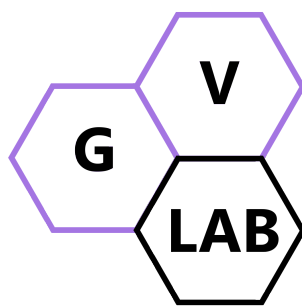
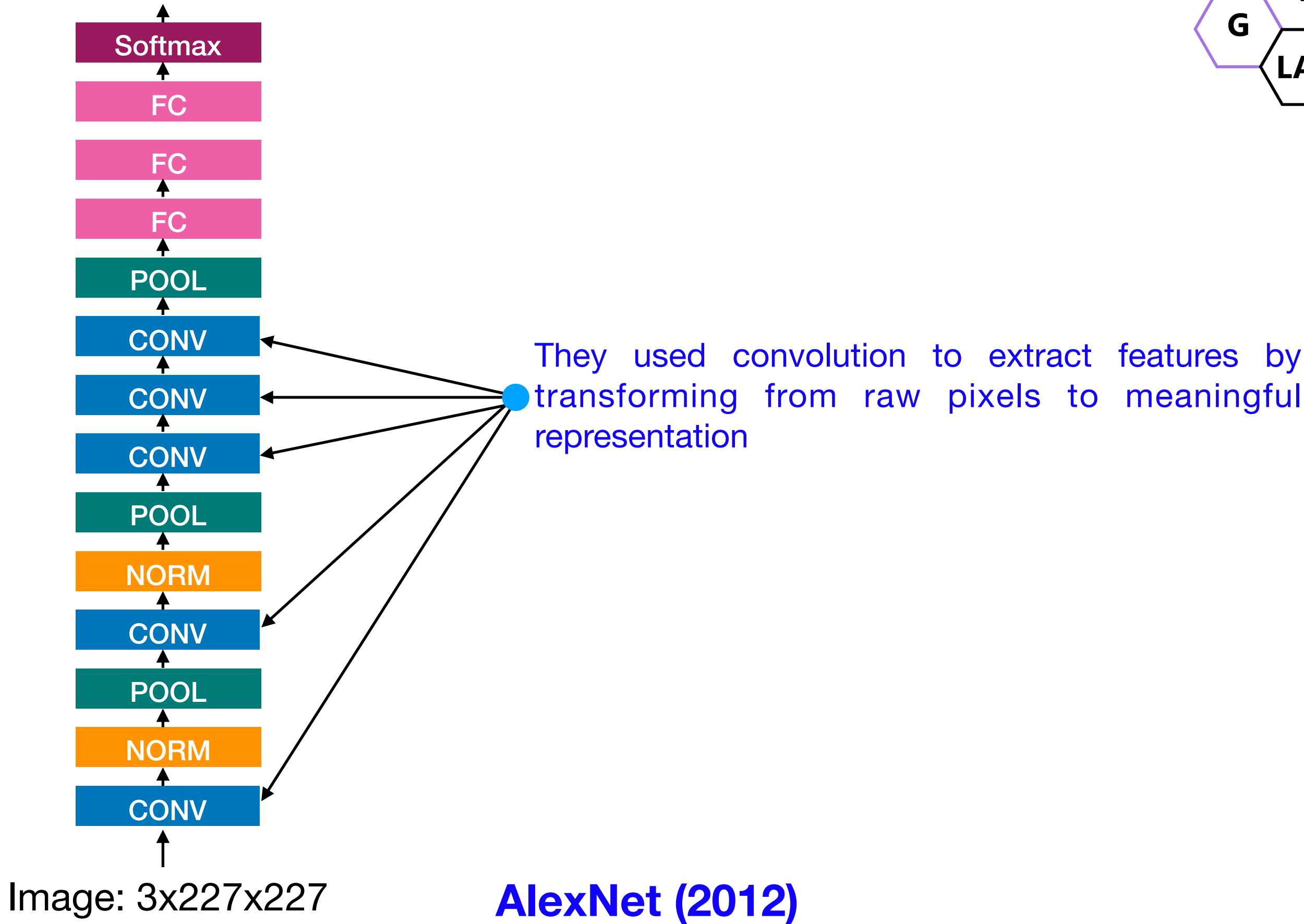
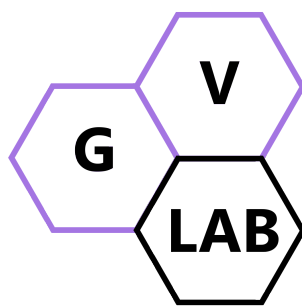


Image: 3x227x227

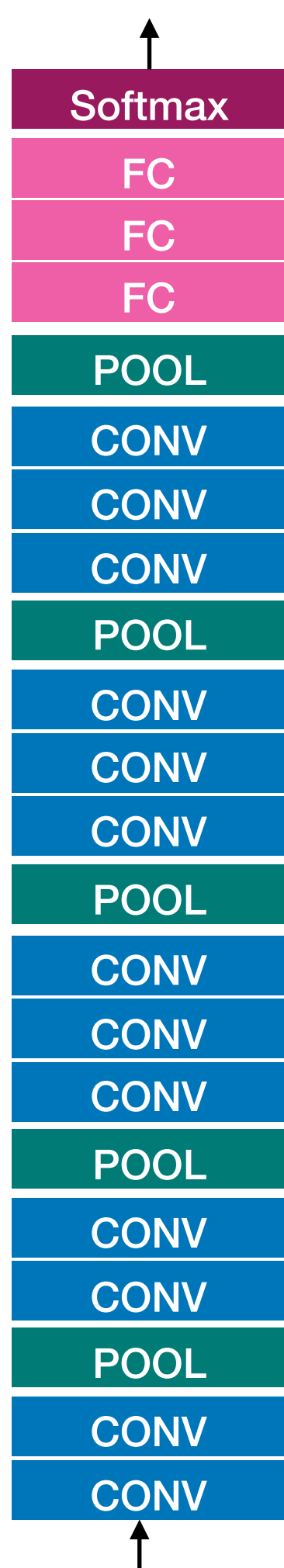
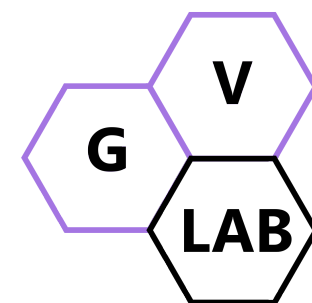
AlexNet (2012)

Alex Krizhevsky and Sutskever, Ilya and Hinton, Geoffrey E, “**ImageNet Classification with Deep Convolutional Neural Networks**,” in Advances in Neural Information Processing Systems, pp.1097-1105, 2012

discrete distribution for 1000 classes



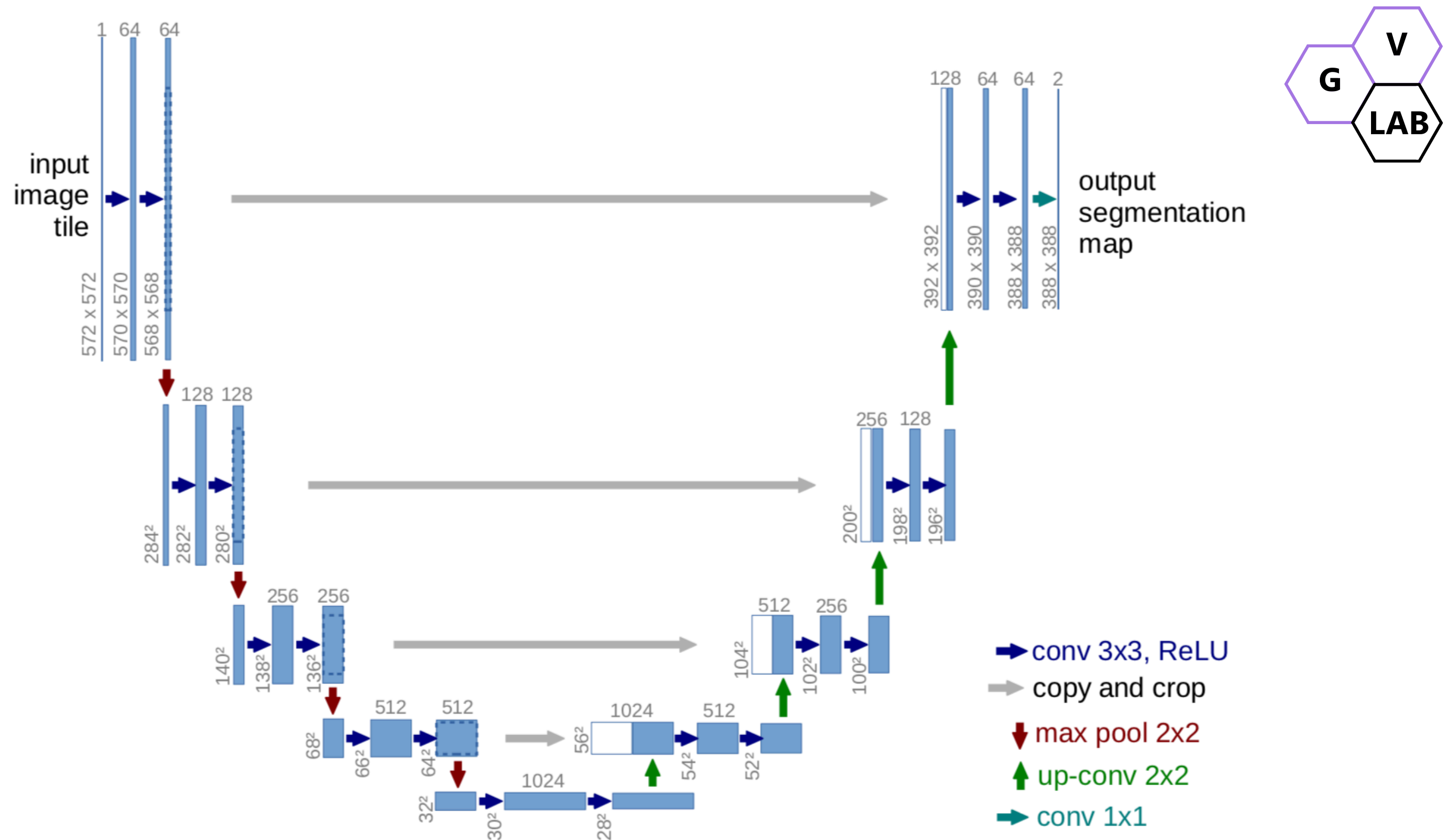
Alex Krizhevsky and Sutskever, Ilya and Hinton, Geoffrey E, "ImageNet Classification with Deep Convolutional Neural Networks," in Advances in Neural Information Processing Systems, pp.1097-1105, 2012



They used convolution to extract features by transforming from raw pixels to meaningful representation

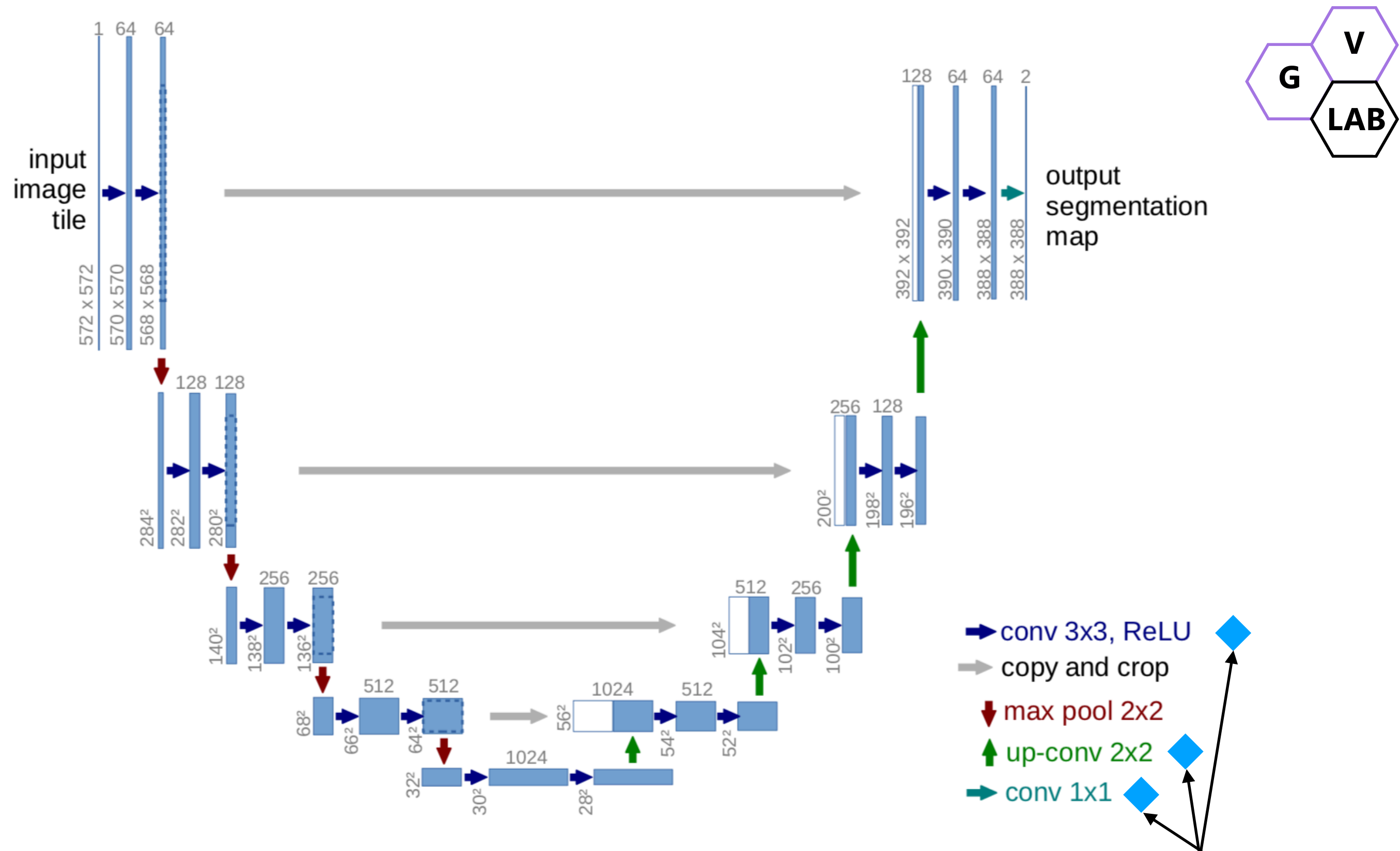
VGG-16 (2014)

AKaren Simonyan, Andrew Zisserman, “**Very Deep Convolutional Networks for Large-Scale Image Recognition**,” arXiv:1409.1556v6



UNet (2015)

Olaf Ronneberger, Philipp Fischer, Thomas Brox, “**U-Net: Convolutional Networks for Biomedical Image Segmentation**,” arXiv:1505.04597v1 [cs.CV]



They used convolution to extract features by transforming from raw pixels to meaningful representation

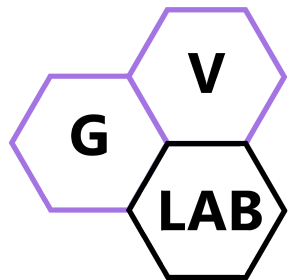
UNet (2015)

Olaf Ronneberger, Philipp Fischer, Thomas Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” arXiv:1505.04597v1 [cs.CV]

Convolution and Cross-correlation

Mathematical Definition

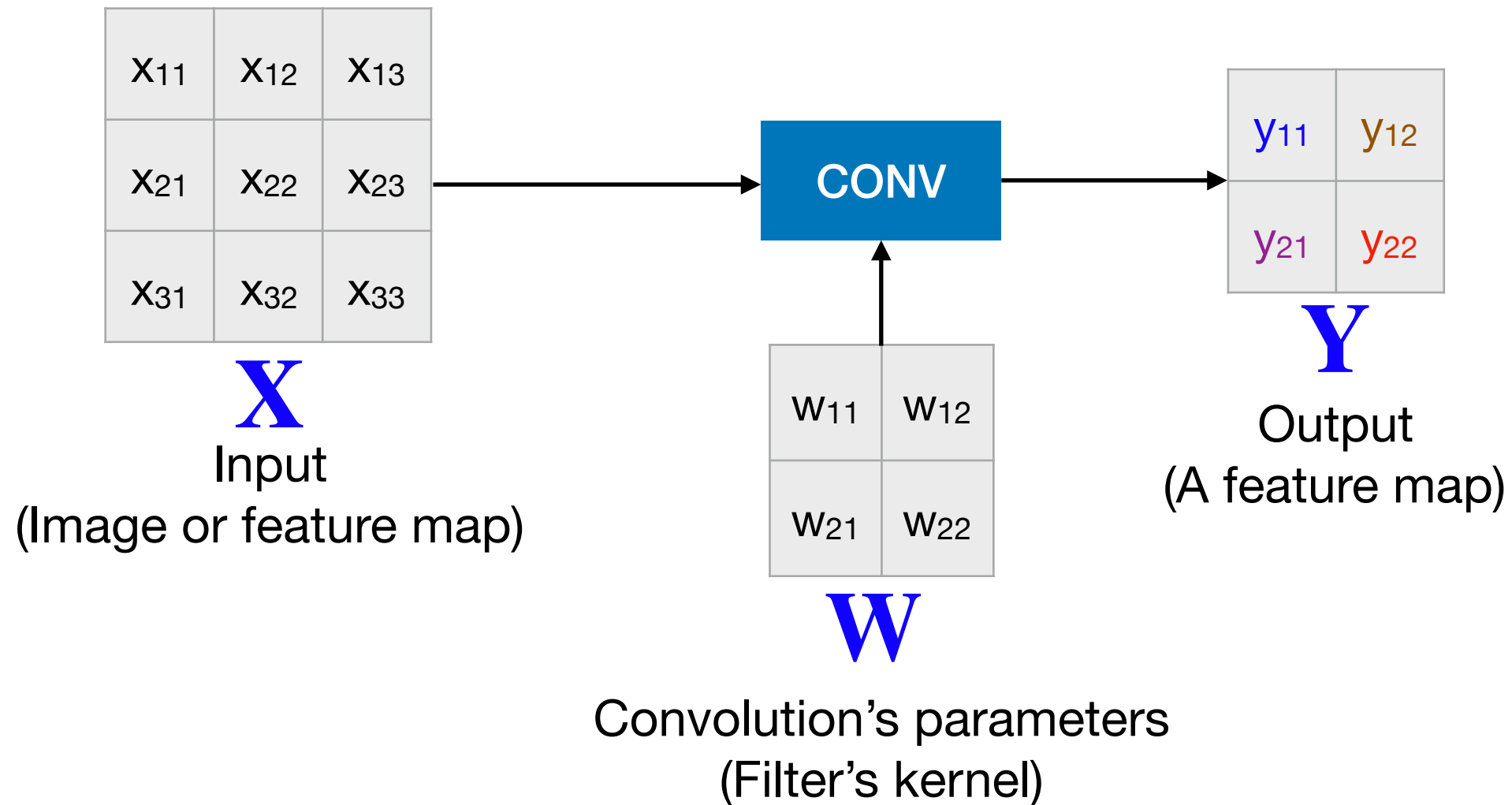
Dr. Thanh-Sach LE
LTSACH@hcmut.edu.vn



GVLab:
Graphics and Vision Laboratory

Faculty of Computer Science and Engineering,
HCMUT

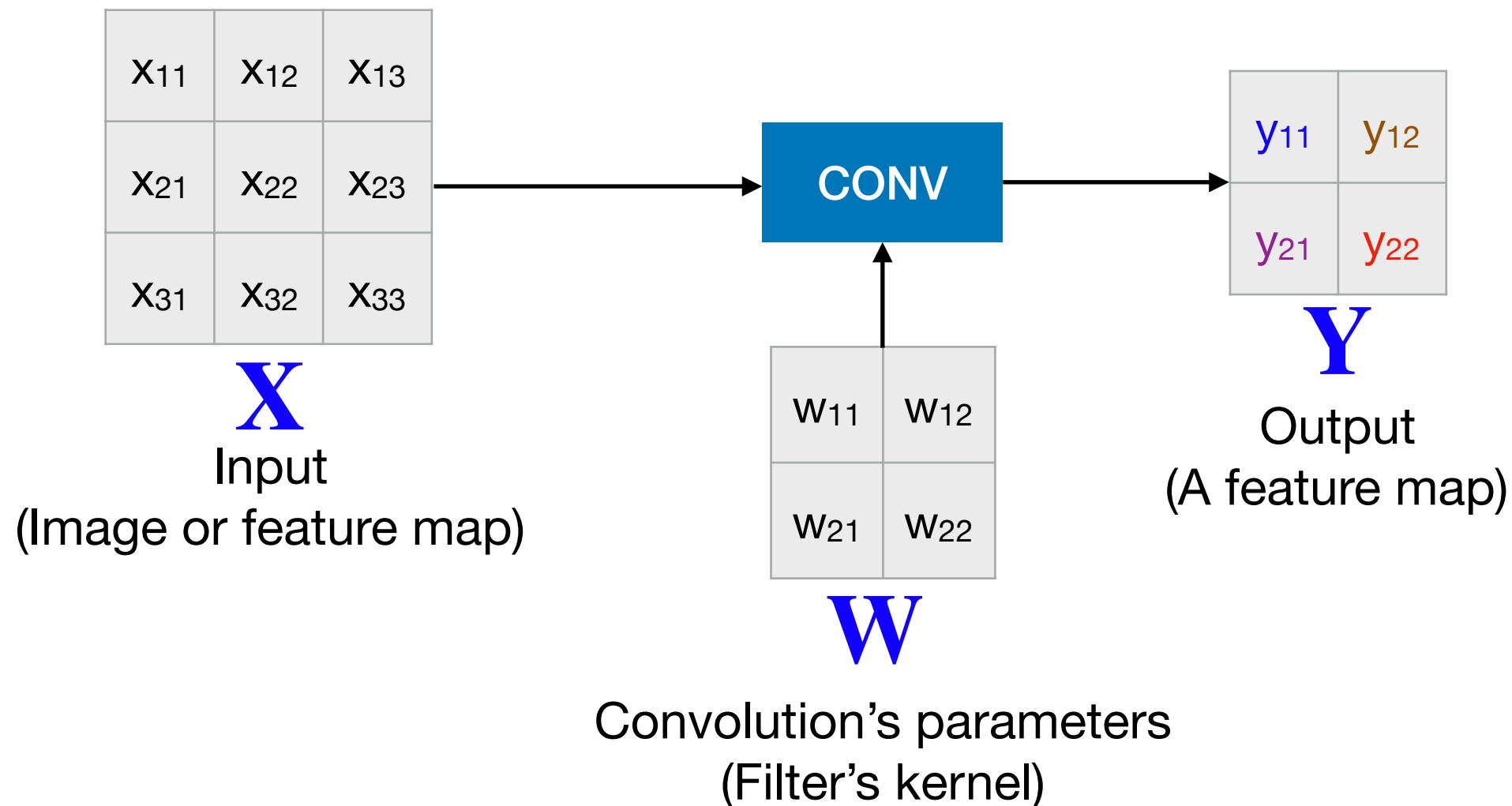
❖ Computation node:



❖ Notation:

$$Y = X * W$$

❖ Computation node:

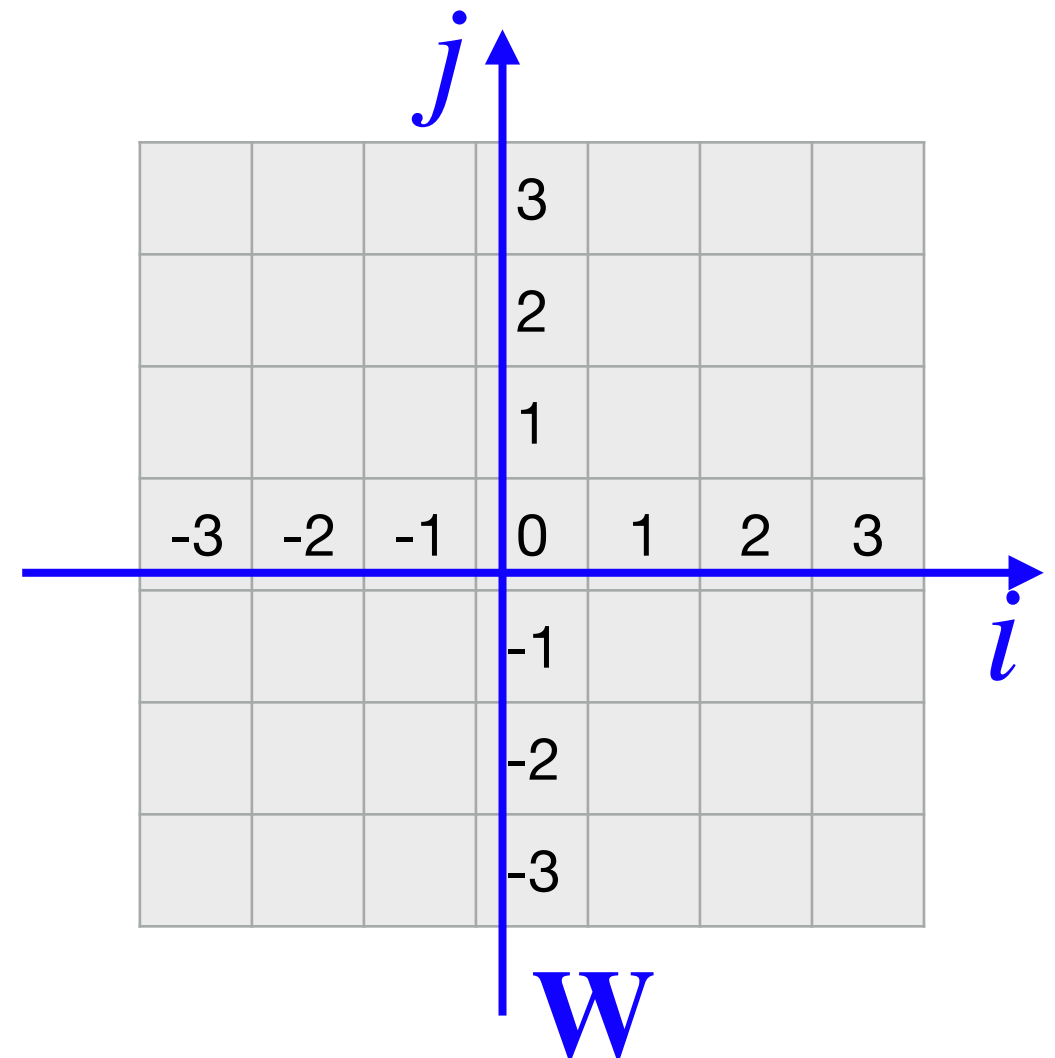
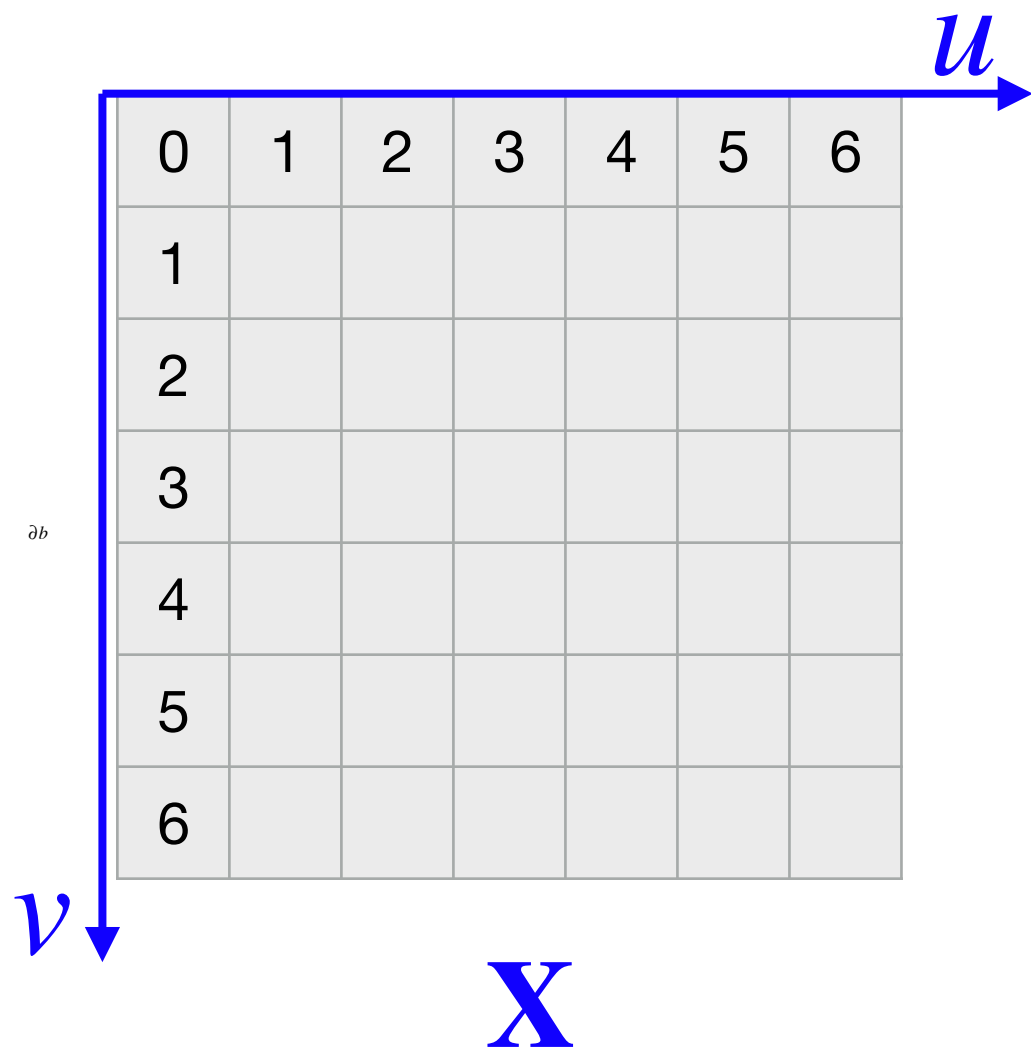


❖ Notation:

$$Y = X * W$$

*** : NOT a matrix multiplication**

❖ Definition:



Radius of kernel:

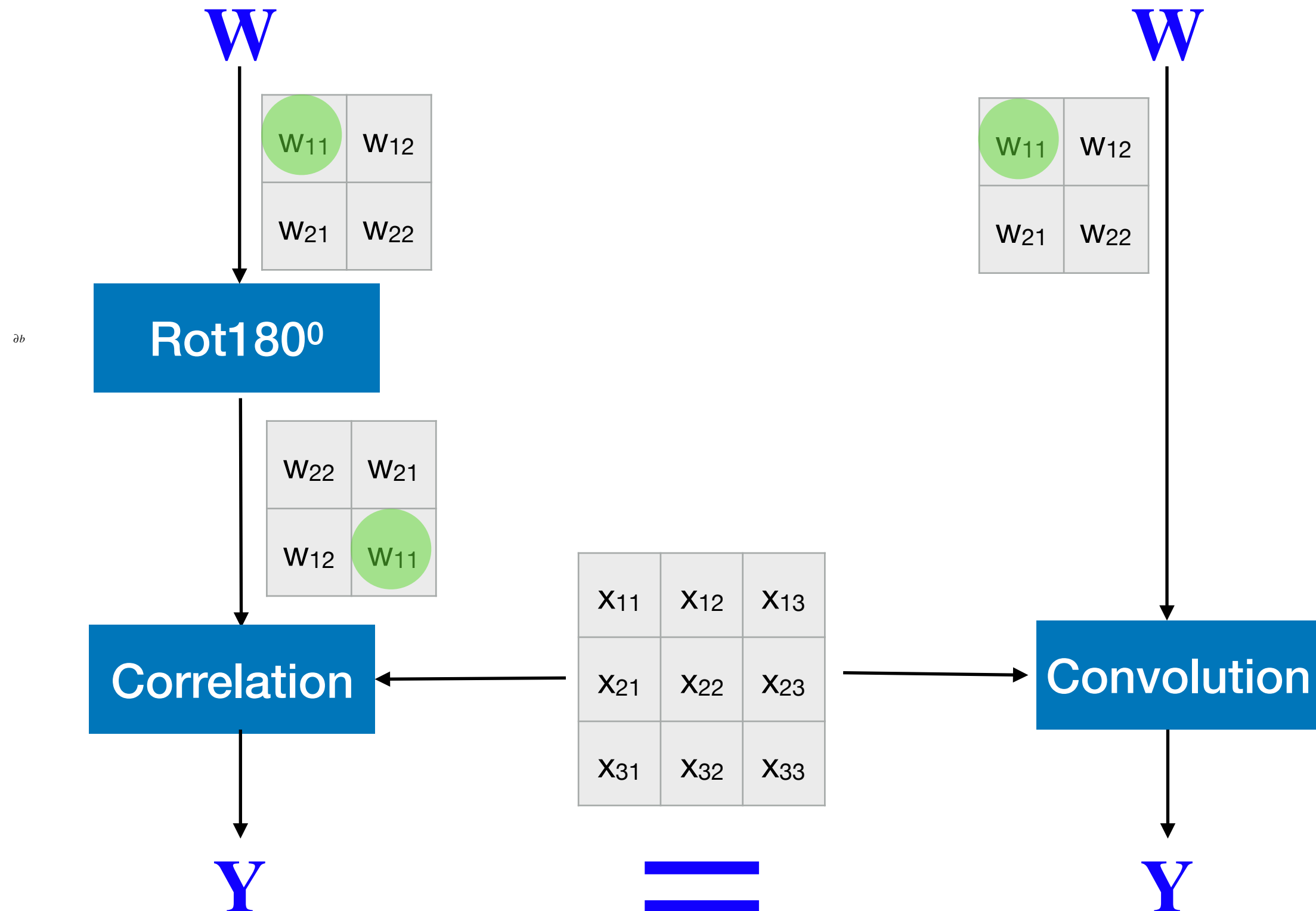
$$W : 7 \times 7 \Rightarrow r = \left\lfloor \frac{7}{2} \right\rfloor = 3$$

❖ Definition:

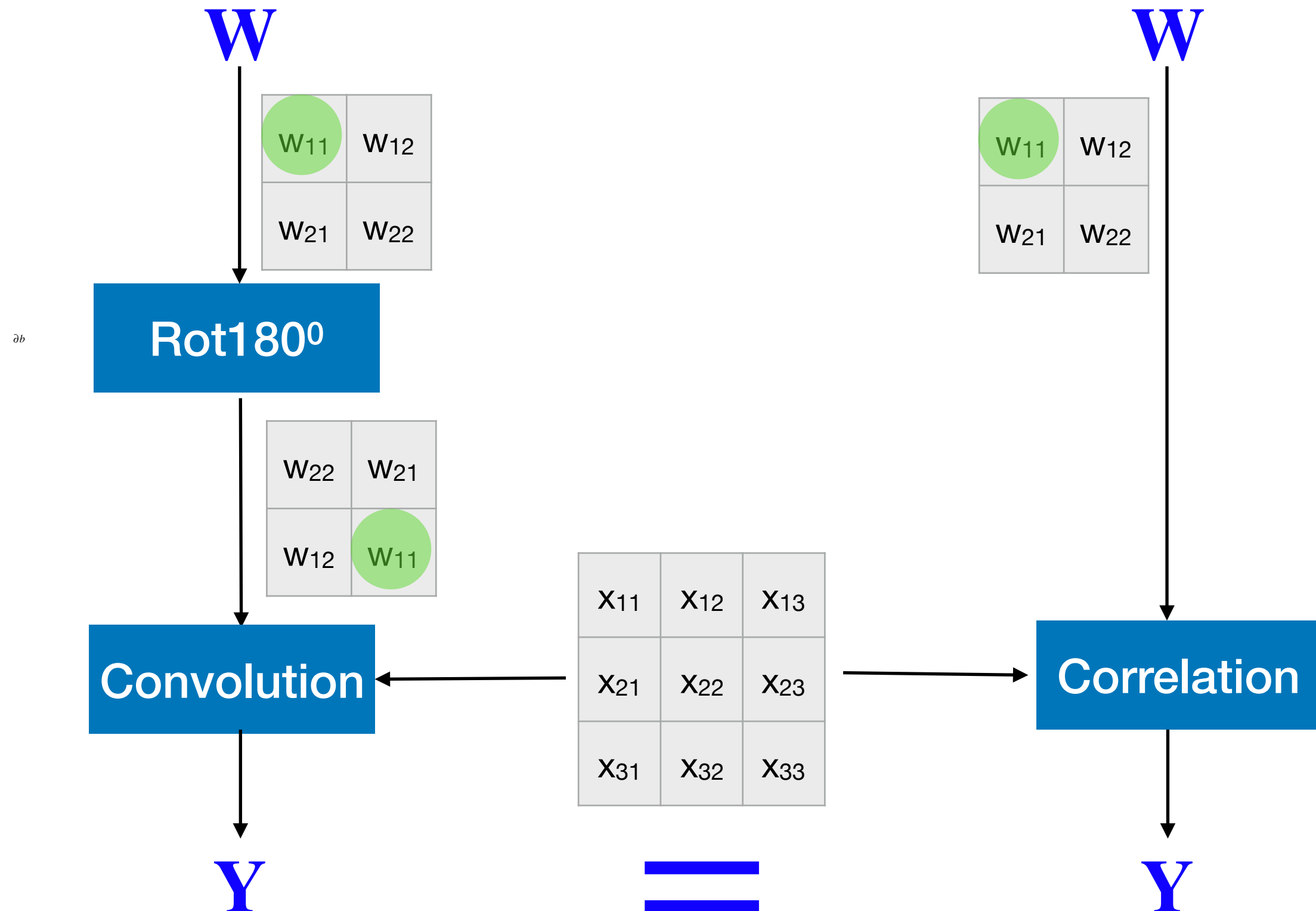
Convolution $\left\{ \begin{array}{l} Y(u, v) = X * W \\ = \sum_{i=-r}^r \sum_{j=-r}^r X(u - i, v - j) W(i, j) \end{array} \right.$

Cross-Correlation $\left\{ \begin{array}{l} Y(u, v) = X \star W \\ = \sum_{i=-r}^r \sum_{j=-r}^r X(u + i, v + j) W(i, j) \end{array} \right.$

❖ Convolution vs cross-correlation:



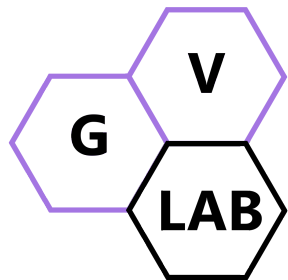
❖ Convolution vs cross-correlation:



Convolution and Cross-correlation

Computation of convolution

Dr. Thanh-Sach LE
LTSACH@hcmut.edu.vn



GVLab:
Graphics and Vision Laboratory

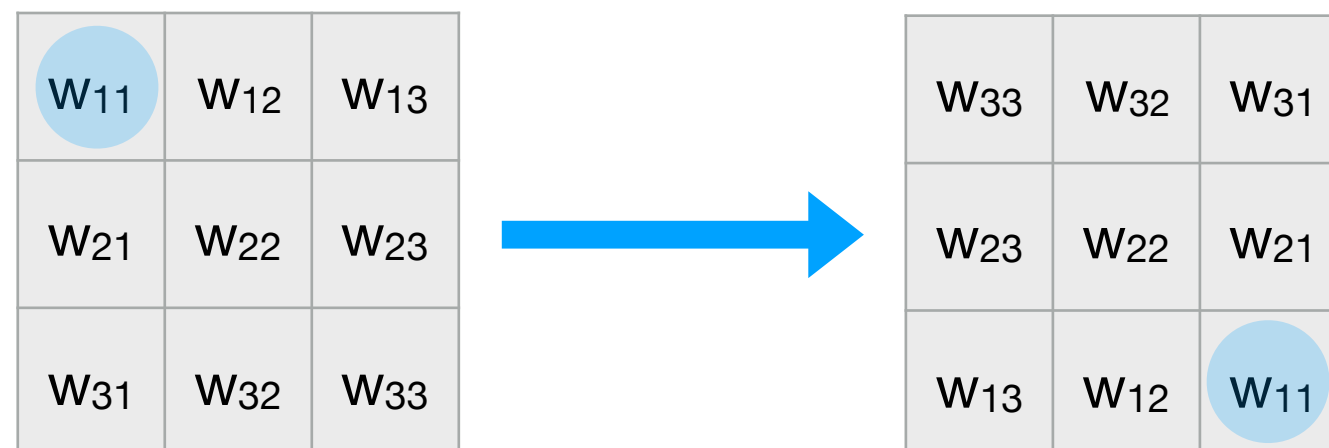
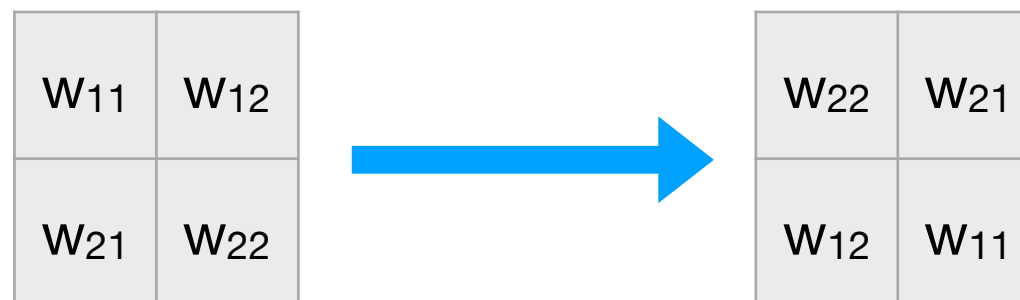
Faculty of Computer Science and Engineering,
HCMUT

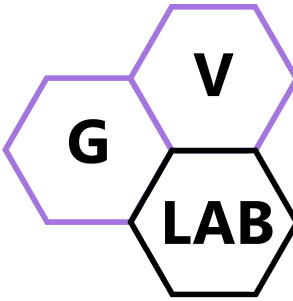
- How to perform the computation (Logical view):

Step 1:

- Flip the kernel around the x-axis and then around the y-axis
- Or, rotate the kernel 180° around its center

∂b





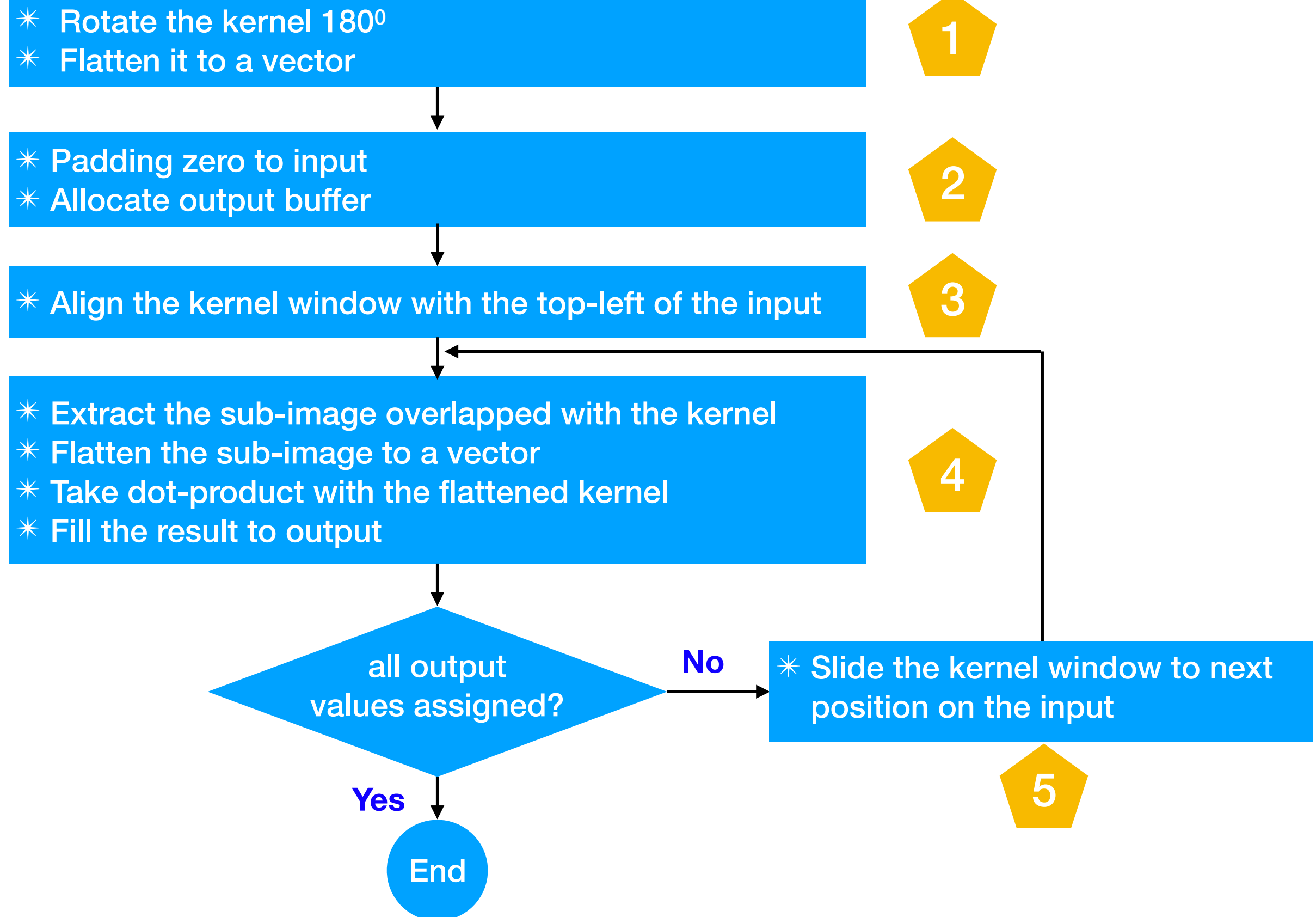
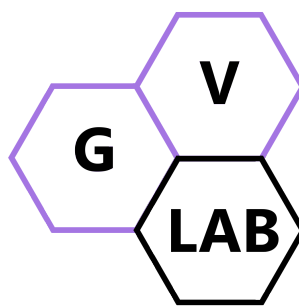
- How to perform the computation (Logical view):

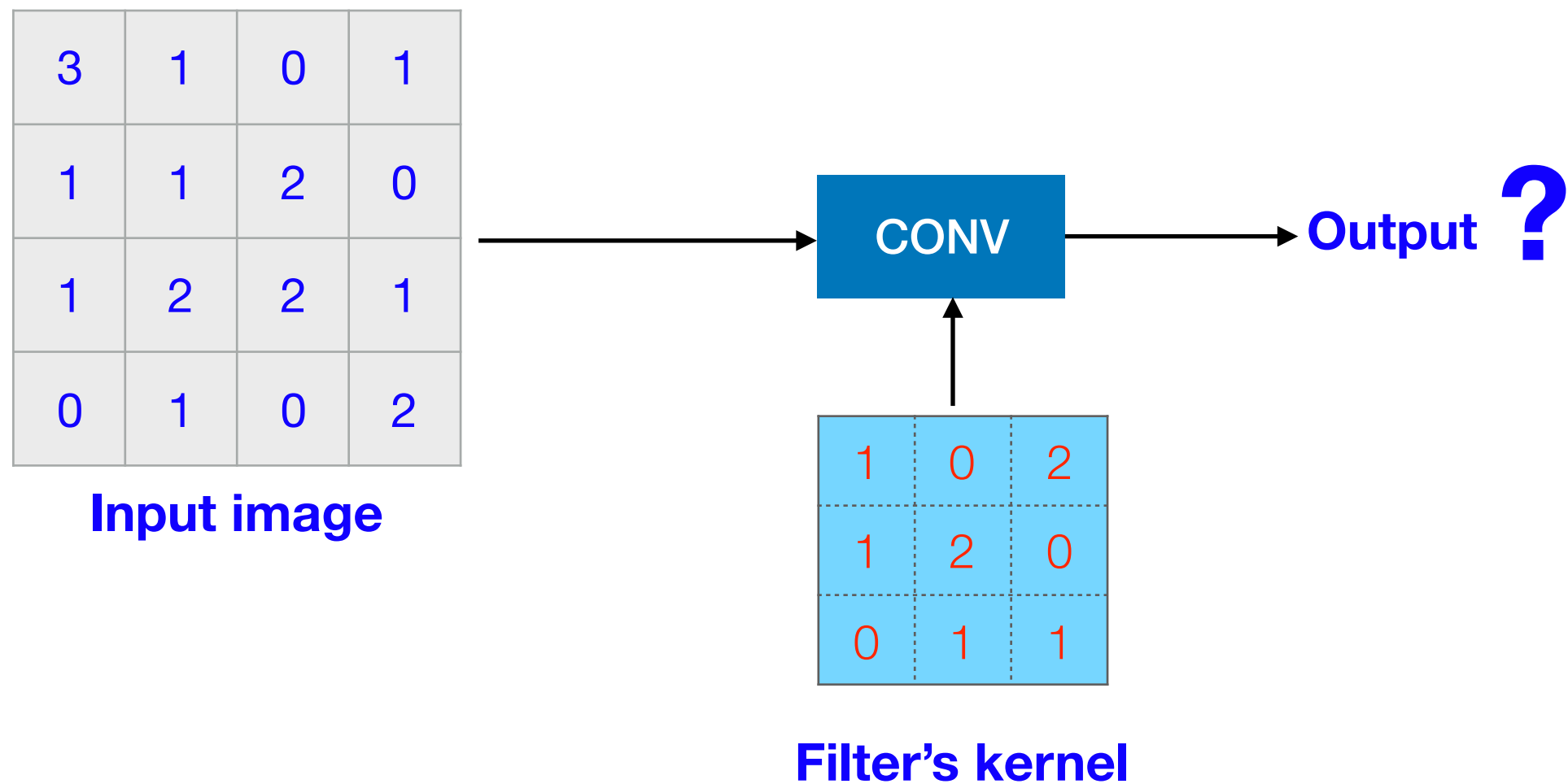
Step 2:

- Compute the cross-correlation between the kernel (obtained from Step 1) with the input image, by:
 - (b) Place the kernel aligned with the left-top of the image
 - (c) Take the **dot product** between the kernel and the sub-image occupied by the kernel and assign the result to the output at the corresponding location
 - (d) Slide the kernel to left and down; do task (b) after each sliding

See the following illustration for detail

Convolution algorithm





1

Rotate the kernel

1	0	2
1	2	0
0	1	1

 W **Rotation 180°**

1	1	0
0	2	1
2	0	1

 $\text{Rot180}^0(W)$

$\text{Rot180}^0(W) = \text{Flip on horizontal direction} + \text{Flip on vertical direction}$

1

Flatten the rotated kernel

1	0	2
1	2	0
0	1	1

 W

Rotation 180°

1	1	0
0	2	1
2	0	1

 $\text{Rot180}^0(W)$

Flattening

1	1	0	0	2	1	2	0	1
---	---	---	---	---	---	---	---	---

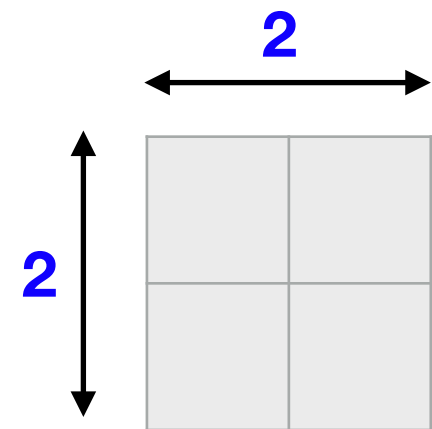
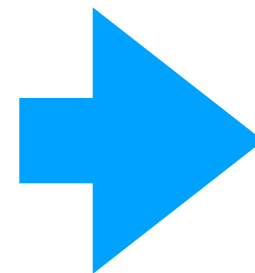
2

Padding the input

3	1	0	1
1	1	2	0
1	2	2	1
0	1	0	2

X

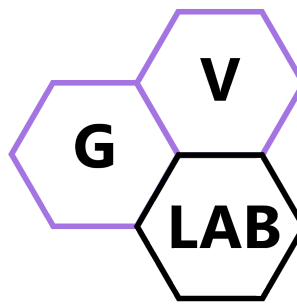
This section illustrates convolution with no padding, stride = 1

**Y**
Output

Output size: 2x2 will be clear shortly

25

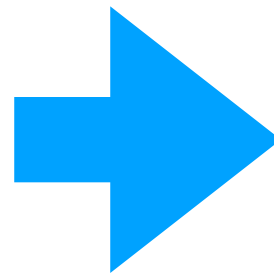
Computation of convolution



3 Align the rotated kernel with the input image

3	1	0	1
1	1	2	0
1	2	2	1
0	1	0	2

Input image



1x3	1x1	0x0	1
0x1	2x1	1x2	0
2x1	0x2	1x2	1
0	1	0	2

1	1	0
0	2	1
2	0	1

$\text{Rot}180^0(W)$

4

Compute dot-product

X and W: aligned at left-top

Get sub-image

1x3	1x1	0x0	1
0x1	2x1	1x2	0
2x1	0x2	1x2	1
0	1	0	2

Input image

1	1	0
0	2	1
2	0	1

 $\text{Rot180}^0(W)$

3	1	0
1	1	2
1	2	2

flattening

3	1	0	1	1	2	1	2	2
---	---	---	---	---	---	---	---	---

dot-product

1	1	0	0	2	1	2	0	1
---	---	---	---	---	---	---	---	---

$$\begin{aligned}
 &3 \times 1 + 1 \times 1 + 0 \times 0 + \\
 &1 \times 0 + 1 \times 2 + 2 \times 1 + \\
 &1 \times 2 + 2 \times 0 + 2 \times 1 \\
 &= 12
 \end{aligned}$$

12	

4

Compute dot-product

X and W: aligned at left-top

Get sub-image

3	1	0	1
1	1	2	0
1	2	2	1
0	1	0	2

Input image

1	0	1
1	2	0
2	2	1

flattening

1	0	1	1	2	0	2	2	1
---	---	---	---	---	---	---	---	---

dot-product

1	1	0	0	2	1	2	0	1
---	---	---	---	---	---	---	---	---

$$\begin{aligned}
 &1 \times 1 + 0 \times 1 + 1 \times 0 + \\
 &1 \times 0 + 2 \times 2 + 0 \times 1 + \\
 &2 \times 2 + 2 \times 0 + 1 \times 1 \\
 &= 10
 \end{aligned}$$

1	1	0
0	2	1
2	0	1

 $\text{Rot180}^0(W)$

12	10

4

Compute dot-product

X and W: aligned at left-top

Get sub-image

3	1	0	1
1	1	2	0
1	2	2	1
0	1	0	2

Input image

1	1	2
1	2	2
0	1	0

flattening

1	1	2	1	2	2	0	1	0
---	---	---	---	---	---	---	---	---

dot-product

1	1	0	0	2	1	2	0	1
---	---	---	---	---	---	---	---	---

$$\begin{aligned}
 &1 \times 1 + 1 \times 1 + 2 \times 0 + \\
 &1 \times 0 + 2 \times 2 + 2 \times 1 + \\
 &0 \times 2 + 1 \times 0 + 0 \times 1 \\
 &= 8
 \end{aligned}$$

1	1	0
0	2	1
2	0	1

 $\text{Rot180}^0(W)$

12	10
8	

4

Compute dot-product

X and W: aligned at left-top

Get sub-image

3	1	0	1
1	1	2	0
1	2	2	1
0	1	0	2

Input image

1	2	0
2	2	1
1	0	2

flattening

1	2	0	2	2	1	1	0	2
---	---	---	---	---	---	---	---	---

dot-product

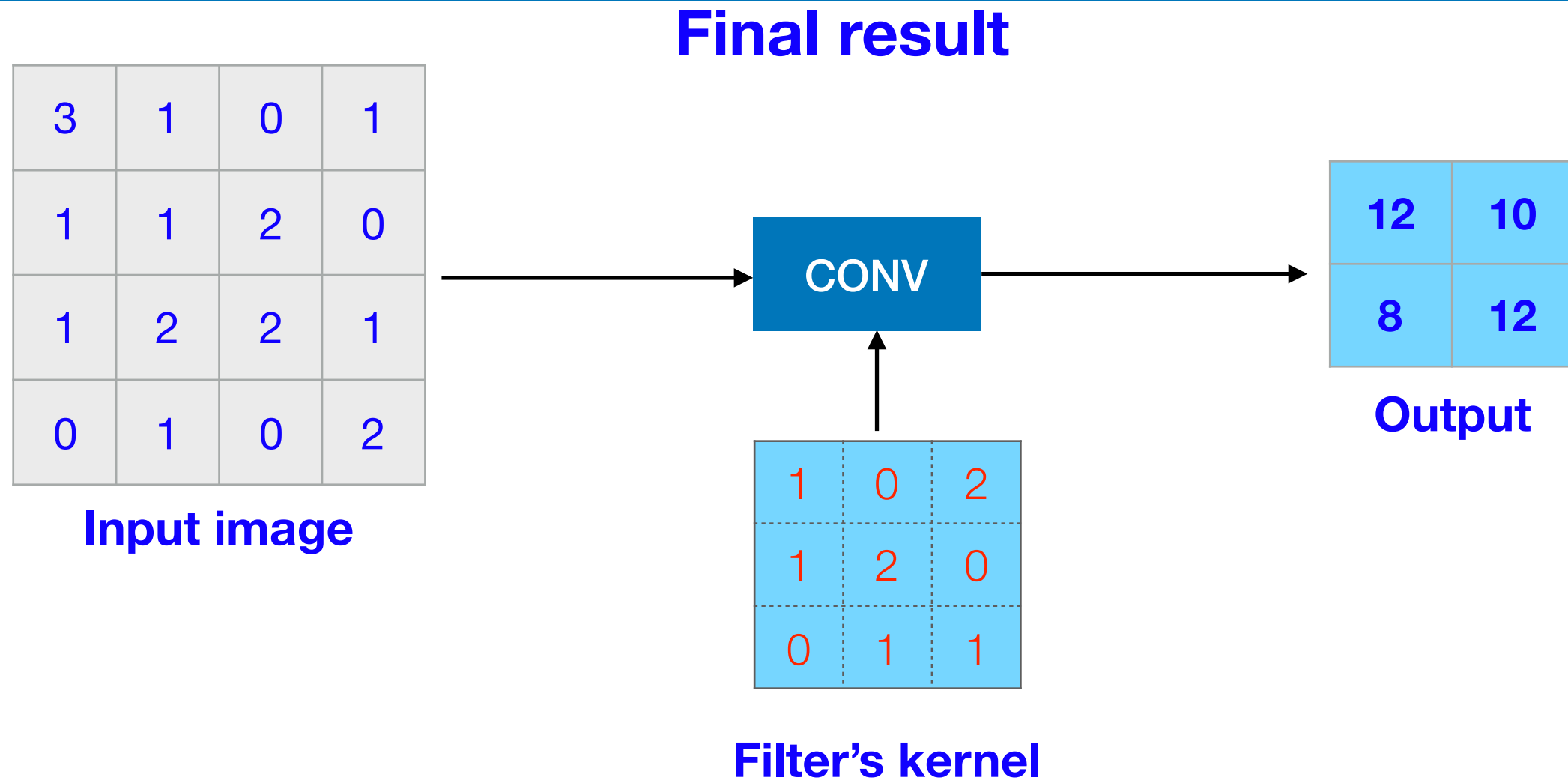
1	1	0	0	2	1	2	0	1
---	---	---	---	---	---	---	---	---

$$\begin{aligned}
 &1 \times 1 + 2 \times 1 + 0 \times 0 + \\
 &2 \times 0 + 2 \times 2 + 1 \times 1 + \\
 &1 \times 2 + 0 \times 0 + 2 \times 1 \\
 &= 12
 \end{aligned}$$

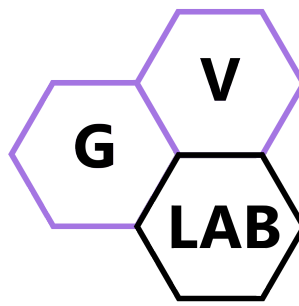
1	1	0
0	2	1
2	0	1

 $\text{Rot}180^0(W)$

12	10
8	12



31 Computation of convolution



Final result

3	1	0	1
1	1	2	0
1	2	2	1
0	1	0	2

Input image
 4×4

CONV

1	0	2
1	2	0
0	1	1

Filter's kernel
 3×3

12	10
8	12

Output

$$(4 - 3 + 1) \times (4 - 3 + 1) \\ 2 \times 2$$

Final result

3	1	0	1
1	1	2	0
1	2	2	1
0	1	0	2

Input image
 4×4

CONV

1	0	2
1	2	0
0	1	1

Filter's kernel
 3×3

12	10
8	12

Output

$$(4 - 3 + 1) \times (4 - 3 + 1) \\ 2 \times 2$$

